1 **(a)** 55 **[B1]**

(b) Angle at the centre is twice the angle at the circumference. **[B1]**

(c) 30 **[B1]**

(d) $(180 - 110) \div 2$ **[M1]**

35 **[A1]**

[Total Marks: 5]

2 **(a)** **(i)** 42 **[B1]**

(ii) 55 **[B1]**

(b) $90 - 42$ **[M1]**

48 **[A1]**

(c) $180 - (55 + 42)$ **[M1]**

83 **[A1]**

[Total Marks: 6]

3 **(a)** **(i)** 62 **[B1]**

(ii) Angle in a semicircle (is 90°) **[B1]**

(b) 28 **[B1]**

(c) Angle $ABC = 62°$ OR Angle $ADB = 90°$ **[M1]**

118 **[A1]**

[Total Marks: 5]

4 **(a)** Angle $ACB = 62$ **[B1]**

Alternate segment theorem **[B1]**

(b) $180 - 62$ **[M1]**

118 **[A1]**

(c) Angle $OAB = 90 - 62 = 28°$ **[M1]**

28 (since triangle OAB is isosceles) **[A1]**

[Total Marks: 6]

5 **(a)** **(i)** Angle $OAT = 90°$ and angle $OBT = 90°$ **[B1]**

Angle $AOB = 180 - x$ **[M1]**

$180 - x = 2(2x - 30)$ **[M1]**

$180 - x = 4x - 60 \implies 5x = 240 \implies x = 48$ **[A1]**

(ii) Angle $AOB = 180 - 48 = 132°$ **[M1]**

Angle $OAB = (180 - 132) \div 2 = 24$ **[A1]**

(b) Angle $ACB = 2(48) - 30 = 66°$ **[M1]**

Angle $ADB = 180 - 66 = 114$ (opposite angles of cyclic quad) **[A1]**

[Total Marks: 8]

6 **(a)** Angle TAB = angle ACB (alternate segment theorem) **[M1]**

$2y - 10 = 3y - 40$ **[M1]**

30 **[A1]**

(b) $x + 10 = 180 - 2(50)$ **[M1]**

70 **[A1]**

(c) Angle $ADB = 180 - 50$ (opposite angles of cyclic quadrilateral) **[M1]**

130 **[A1]**

[Total Marks: 7]